# Weak solution of the merged mathematical equations of the polluted atmosphere

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**Abstract**

Considered as a geophysical fluid, the polluted atmosphere shares the shallow domain characteristics with other natural large-scale fluids such as seas and oceans. This means that its domain is excessively greater horizontally than in the vertical dimension, leading to the classic hydrostatic approximation of the Navier-Stokes equations. The authors of the [Azérad and Guillén, 2001] article have proved a convergence theorem for this model with respect to the ocean, without considering pollution effects. In this present work we provide a generalization of their result translated to the atmosphere, extending the fluid velocity equations with an additional convection-diffusion equation representing pollutants in the atmosphere.

**Keywords:** Navier-Stokes equations, shallow domains, pollution evolution equation, hydrostatic approximation, compactness, weak solutions.

# Contents

# 1 Introduction

We investigate the convergence of the weak solutions for the merged motion-pollution atmosphere model. At this point the functions we work with are $\mathbf{v} = (\mathbf{v}_H, v_3)$, and $C$, representing velocity and concentration, respectively (we are not considering other variables such as water vapor quantity or temperature).

We underline the fact that this work is for the case of **inland** scenario.

Even though the [Azérad and Guillén, 2001] article works with the equations of the ocean, this work is the extension of that article in the sense that we transfer and exploit its main result for the case of the atmosphere, and for this reason it is the main reference article for this work.

This paper is organized in the following way. In section 2 we describe the physical model of the polluted atmosphere and the scaling leading to the hydrostatic equations. The main theorem is stated in section 3. In section 4 we describe the function spaces we work in and present the weak formulation. The main theorem's proof is described in section 5, and a generalized model is introduced in chapter 6. A few concluding remarks can be found in section 7, finally in the appendix of the article we present an existence result for the hydrostatic equations of the polluted atmosphere.

## 1.1 Notations

a) $\epsilon =$ the aspect ratio which goes to zero

b) $\nu =$ viscosity

c) $\Omega_\epsilon, \Omega =$ the original and the rescaled, $\epsilon$-independent domain, respectively

d) $g =$ represents the force due to gravity, which also includes the centrifugal effect

e) $\mathbf{w}$ represents the earth rotation angular speed, $l(y) =$ latitude

f) $\alpha = 2f\sin(l(x_2))$, $\beta = 2f\cos(l(x_2))$

g) $(\cdot,\cdot) =$ the scalar product in $L^2(\Omega)^d$ or the duality $L^p(\Omega)$, $L^{p'}(\Omega)$

h) $< \cdot,\cdot >_\Gamma =$ the scalar product or duality on the boundary curve $\Gamma$ ($H^{-1/2}(\Gamma)$, $H^{1/2}(\Gamma)$)

i) $b(\mathbf{u}_H) = \alpha(-u_2, u_1)$

# 2 The equations describing the polluted atmosphere

The first step is to choose a coordinate system, set the domain we start with and an appropriate set of equations describing both atmosphere motions and the effect of pollution. The set of equations, the choice of the locally acceptable

Cartesian coordinate system we are using for the velocity terms are similar to those suggested by our reference article. We ignore large scale effects such as the curvature of the Earth. For describing the effect of pollution we use a continuity equation written in Cartesian coordinates for the concentration.

We fix the local Cartesian coordinate system to be adjusted to the downwind direction, i.e. $z$ is the vertical direction, while $x$ is the downwind, and $y$ is the crosswind direction. It is a legitimate thing to do if we are considering a time interval which is not too long in the sense that it is reasonable to assume that we have a permanent, relatively stable main wind direction. It is true that this is a certain restriction on the model, but as we will see later, exploiting this choice will help us simplify the model a bit.

Let us consider the following domain to begin with. This domain is a local slice of the atmosphere filled with homogeneous air that we assume to be incompressible, and a pollutant of concentration $C$.

$$\Omega_\epsilon = \{(x, y, z) \in \mathbb{R}^3; (x, y) \in \omega, 0 < z < \epsilon h(x, y)\}, \tag{1}$$

where $\omega$ is a Lipschitz-domain in $\mathbb{R}^2$, and $h$ is a nonnegative Lipschitzian application.

The equations describing the air velocity in the atmosphere are the incompressible Navier-Stokes equations:

$$\partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla)\mathbf{v} - \Delta_\nu \mathbf{v} + \nabla q + 2\mathbf{w} \times \mathbf{v} = \mathbf{g} \text{ in } \Omega_\epsilon \times (0, T) \tag{2}$$

$$\nabla \cdot \mathbf{v} = 0 \text{ in } \Omega_\epsilon \times (0, T) \tag{3}$$

Contaminants emitted into the atmosphere are transported by the wind, mixed and dispersed by turbulent behavior, and deposited onto the ground by gravity. The dispersion of pollutants is commonly modeled by a continuity equation, and this is what we will use to describe the pollution phenomena as well.

$$\partial_t C + \mathbf{v} \cdot \nabla C = K_x \frac{\partial^2 C}{\partial x^2} + K_y \frac{\partial^2 C}{\partial y^2} + K_z \frac{\partial^2 C}{\partial z^2} + S \tag{4}$$

In this equation $S$ represents the sources and sinks of the pollutant in the atmosphere i.e. its emission, removal from the atmosphere by dry and wet deposition, and chemical reactions. We will use the approach suggested by [Zhuk et al., 2016] and use what is called a parametrized nascent delta function (a Gaussian impulse, specifically) to describe the source term $S$.

Let us consider a gas plume with intensity $I$ originating at emission coordinates $\mathbf{x}_s$ beginning at time $t_s$. At the beginning we assume zero concentration. An abrupt gas emission is modeled by a source term represented as a product of an approximate delta function $\delta_\kappa(\mathbf{x} - \mathbf{x}_s)$ (centered at $\mathbf{x}_s$ with a constant parameter $\kappa$) and a Heaviside function $s$ (switching from 0 to $I$ at time $t_s$.) In fact, this source term models the continuous emission of the gas at constant rate and fixed location. Basically $s(t; t_s)$ is a function which activates the source, namely $s(t;t_s) := IH(t-t_s)$ where $H(\cdot)$ is the Heaviside step function and $0 \le t_s \le T$. We arrive to the complete source term model of the form $S_\kappa(t,t_s,x,x_s) = s(t;t_s)\delta_\kappa(\mathbf{x}-\mathbf{x_s})$, while in the limit model the source term $S$ naturally stands for the $\delta$ distribution.

In the following we will describe a usual practice in asymptotic analysis, namely we perform a vertical scaling to make the domain independent of $\epsilon$.

$$x = x_1, y = x_2, z = \epsilon x_3$$
$$v_1 = u_1^\epsilon, v_2 = u_2^\epsilon, v_3 = \epsilon u_3^\epsilon, p = p^\epsilon$$
$$\nu_x = \nu_1, \nu_y = \nu_2, \nu_z = \epsilon^2 \nu_3$$
$$K_z = \epsilon^2 K_3$$

So we have the following new domain

$$\Omega = \{(x_1,x_2,x_3) \in \mathbb{R}^3; (x_1,x_2) \in \omega, 0 < x_3 < h(x_1,x_2)\}.$$

As a second step of the rescaling process, here we also exploit the fact that we chose the coordinate system in a way so that the $x$ axis matches the downwind direction. In this case we use an additional scaling equation, which is different in nature from the previous ones in the sense that it is not derived from the shallowness of the domain but rather suggested by the fact that in the downwind direction it is natural to assume that the diffusion is negligible compared to downwind transport, i.e.

$$|u_1 \partial_{x_1} C| \gg \left| K_x \frac{\partial^2 C}{\partial x_1^2} \right| \tag{5}$$

which leads us to using

$$K_x = \epsilon K_1.$$

Finally, as we mentioned before, we use Gaussian approximate delta functions to represent the source, so it is convenient to involve the $\epsilon$ term in one additional point in the functions, which is the standard deviation parameter of the distribution describing the source — because as later we will take the limit as $\epsilon$ goes to zero, it will result in having the $\delta$ distribution representing the hydrostatic source term, which is a physically natural choice. In other words, we choose $\kappa$ to be $\epsilon$ and the following function will give shape to the source term: $\delta_\epsilon(\mathbf{x}-\mathbf{x_s}) = \gamma\epsilon^{-1} e^{\frac{-(\mathbf{x}-\mathbf{x_s})^2}{\epsilon^2}}$; and finally $S_\epsilon = IH(t-t_s)\gamma\epsilon^{-1}e^{\frac{-(\mathbf{x}-\mathbf{x_s})^2}{\epsilon^2}}$, where $\gamma$ is a normalizing constant.

Putting this condition and the previously described rescaling terms together to the original equations, we arrive to the merged anisotropic equations of the polluted atmosphere above inland surface.

$$\partial_t u_1^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_1^\epsilon - \Delta_\nu u_1^\epsilon - \alpha u_2^\epsilon + \epsilon\beta u_3^\epsilon + \partial_1 p^\epsilon = 0 \text{ in } \Omega \times (0,T) \tag{6}$$

$$\partial_t u_2^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_2^\epsilon - \Delta_\nu u_2^\epsilon + \alpha u_1^\epsilon + \partial_2 p^\epsilon = 0 \text{ in } \Omega \times (0,T) \tag{7}$$

$$\epsilon^2\{\partial_t u_3^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_3^\epsilon - \Delta_\nu u_3^\epsilon\} - \epsilon\beta u_1^\epsilon + \partial_3 p^\epsilon = 0 \text{ in } \Omega \times (0,T) \tag{8}$$

$$\nabla \cdot \mathbf{u}^\epsilon = 0 \text{ in } \Omega \times (0,T) \tag{9}$$

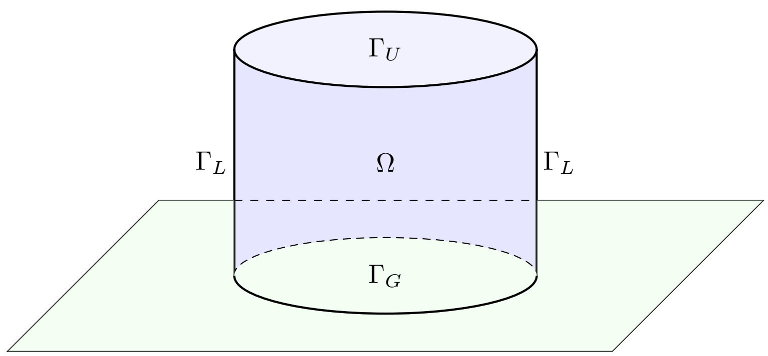
$$\partial_t C^\epsilon + \mathbf{u}^\epsilon \cdot \nabla C^\epsilon = \epsilon K_1 \frac{\partial^2 C^\epsilon}{\partial x_1^2} + K_2 \frac{\partial^2 C^\epsilon}{\partial x_2^2} + K_3 \frac{\partial^2 C^\epsilon}{\partial x_3^2} + S_\epsilon \text{ in } \Omega \times (0,T) \tag{10}$$

$$\mathbf{u}^\epsilon = 0 \text{ on } \Gamma \times (0,T) \tag{11}$$

$$\mathbf{u}^\epsilon(\cdot, t=0) = \mathbf{u}_0, C^\epsilon(\cdot, t=0) = C_0^\epsilon \text{ in } \Omega \tag{12}$$

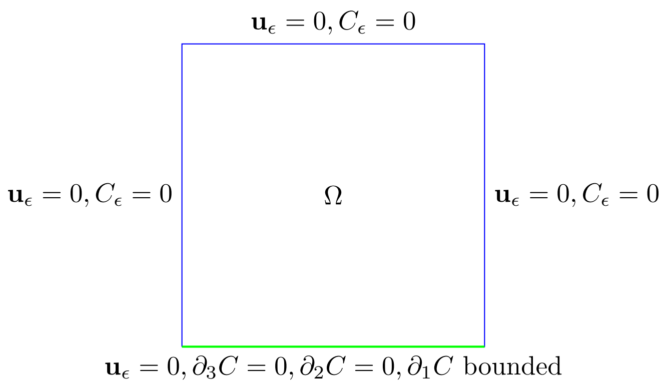
To make the above set of equations complete, we need to decide on the boundary conditions for the pollution term.

It is vital to choose boundary conditions for the pollution concentration that are physically as accurate as possible and at the same time not unnecessarily complicated. As we can see in many other articles and as we will see later in this one, in the case of LAMs (limited area models) finding the right boundary conditions can be a nontrivial task.




The boundary division of the $\partial\Omega$: the upper boundary of the $\Omega$ domain is denoted by $\Gamma_U$, the lateral boundary is $\Gamma_L$, and the lower boundary of the domain is $\Gamma_G$. As we mentioned before, the case we are investigating is that of inland domains $\Omega$, i.e. the case of landscapes and cities with no significant nearby water surface. This is very important since the boundary conditions for both wind velocity and pollution concentration in the air are different depending whether we are above solid ground or water surface. The boundary conditions we choose to use in the inland scenario are the following, shown in a cross-section.

An explanation of the boundary conditions:




- The upper boundary of the domain we are considering is chosen to be at the isobar representing the planetary boundary layer ([Arya, 2001]), on this $p = p_0$ pressure level the upper boundary of the local domain is denoted by $\Gamma_U$. On $\Gamma_U$ we use $\mathbf{u}_H = 0, u_3 = 0, C(x, y, z) = 0$ boundary conditions.

- On $\Gamma_L$ we use the same boundary conditions as on the upper level, i.e. we have $\mathbf{u}_H = 0, u_3 = 0, C(x, y, z) = 0$.

  **Remark.** *If we wanted to use $C(x, y, z) = c$ with some $c$ constant value, we could make a change of variables distracting that background flow and arrive back to the same mathematical problem with $C$ being zero.*

- In this model we neglect topological variations, and the $z = 0$ plane represents the lower boundary. On the ground level we have $u_H = 0, u_3 = 0$ boundary conditions for the air velocity. This is the mathematical way of expressing no-slip boundary conditions for the horizontal wind speed and the impervious nature of the ground with respect to the wind. As for the concentration, the way we can choose ground level boundary conditions is the following. Firstly, we assume that the pollution is not absorbed by the ground, so we use $\partial_3 C = 0$ on $\Gamma_G$ (see also for example [Yeh and Huang, 1975]). For the horizontal partial derivatives of the pollution, we can separate downwind and crosswind direction boundary conditions. In the crosswind direction we use $\partial_2 C = 0$ while in the downwind (stronger) direction our assumption is more relaxed, we only assume that $\partial_1 C$ is finite in the boundary norm, i.e. in $L^2 L^2(0, T; \Gamma_G)$.

Note that we have a domain where the lower and upper boundaries represent a physically existent layer (i.e. the ground and the planetary boundary layer), but outside the lateral boundary the physical world continues without influence and without any particular external condition. On boundaries of such nature we have to apply OBCs (open boundary conditions), which is not a trivial task, see for example [Wang et al., 2014]. We need to avoid any spurious reflection or constraint which is unnatural, but at the same time we want the model to remain mathematically manageable.

If we assume $\mathbf{u}^\epsilon = O(1)$, then neglecting the $\epsilon^2$ and $\epsilon$ in the anisotropic equations, we formally arrive to the hydrostatic Navier-Stokes equations combined with pollution.

$$\partial_t u_1 + \mathbf{u} \cdot \nabla u_1 - \Delta_\nu u_1 - \alpha u_2 + \partial_1 p = 0 \text{ in } \Omega \times (0, T) \tag{13}$$

$$\partial_t u_2 + \mathbf{u} \cdot \nabla u_2 - \Delta_\nu u_2 - \alpha u_1 + \partial_2 p = 0 \text{ in } \Omega \times (0, T) \tag{14}$$

$$\partial_3 p = 0 \text{ in } \Omega \times (0, T) \tag{15}$$

$$\nabla \cdot \mathbf{u} = 0 \text{ in } \Omega \times (0, T) \tag{16}$$

$$\partial_t C + \mathbf{u} \cdot \nabla C = K_2 \frac{\partial^2 C}{\partial x_2^2} + K_3 \frac{\partial^2 C}{\partial x_3^2} + S \text{ in } \Omega \times (0, T) \tag{17}$$

$$u_1 = u_2 = u_3 n_3 = 0 \text{ on } \Gamma \times (0, T) \tag{18}$$

$$u_i(\cdot, t = 0) = u_{0i}, C(\cdot, t = 0) = C_0 \text{ in } \Omega, i = 1{,}2 \tag{19}$$

**Remark.** *Here $S$ stands for the $\delta$ limit distribution. We keep the notation $S$ for generality, since as we will see later, instead of the approximate delta functions and the delta distribution, it is possible to use any sort of "sufficiently smooth" source term (which can be independent from $\epsilon$ in the anisotropic case as well) that is bounded in $L_t^2 L_x^2$.*

# 3 Main theorem

- The article of [Azérad and Guillén, 2001] proved a convergence theorem for weak solutions of the equations describing the ocean without pollution. We can use a modified version of this, translated to the polluted inland atmosphere domain.

  **Theorem 1.** *Let $\mathbf{u}_0 \in L^2(\Omega)^3$, with $\nabla \cdot \mathbf{u}_0 = 0, \mathbf{u}_0 \cdot \mathbf{n} = 0$ on $\partial\Omega$; there exists a weak solution $\mathbf{u}$ of the hydrostatic Navier-Stokes equations as the aspect ratio $\epsilon$ tends to 0.*

- The task we want to do is to extend this result in the sense that it becomes valid to the merged set of $(\mathbf{u}, C)$ equations as well.

  **Theorem 2.** *Let $\mathbf{u}_0 \in L^2(\Omega)^3$, with $\nabla \cdot \mathbf{u}_0 = 0, \mathbf{u}_0 \cdot \mathbf{n} = 0$ on $\partial\Omega$; and let $C(0) \in L^2(\Omega)^3$, $C(0) = 0$ on $\partial\Omega$, there exists a weak solution ($\mathbf{u}$, $C$) of the hydrostatic Navier-Stokes equations as the aspect ratio $\epsilon$ tends to 0.*

# 4 Weak formulation

We need to calculate the weak formulation of the merged equations in both the anisotropic and hydrostatic case. Since this inland atmosphere model has slightly different boundary conditions even in the velocity equations than the model of the ocean, we will see that the weak formulation of our system has minor differences from the ocean scenario.

## 4.1 Weak formulation of the hydrostatic system

For the weak formulation we will use the modified version of the spaces defined in the original article.

$$H^1_\Gamma(\Omega) = \{v \in H^1(\Omega); v = 0 \text{ on } \partial\Omega = \Gamma\}$$

$$\mathbf{V} = \{\mathbf{v} \in H^1_\Gamma(\Omega) \times H^1_\Gamma(\Omega) \times H^1_0(\Omega); \nabla \cdot \mathbf{v} = 0 \text{ in } \Omega\}$$

$$H(\partial_3, \Omega) = \{v \in L^2(\Omega); \partial_3 v \in L^2(\Omega)\}$$

$$H_0(\partial_3, \Omega) = \{v \in H(\partial_3, \Omega); v n_3 = 0 \text{ on } \partial\Omega\}$$

$$\mathbf{W} = \{\mathbf{u} \in H^1_\Gamma(\Omega) \times H^1_\Gamma(\Omega) \times H_0(\partial_3, \Omega); \nabla \cdot \mathbf{u} = 0 \text{ in } \Omega\}$$

Find $(\mathbf{u}, C)$ such that $\mathbf{u} = (\mathbf{u}_H, u_3) \in L^2(0, T; \mathbf{W})$ with $\mathbf{u}_H \in L^\infty(0, T; L^2(\Omega)^2)$, and $C$ with $\partial_2 C, \partial_3 C \in L^2_t L^2_x$, $C \in L^\infty L^2$ and $C(0) \in L^2_x$, such that

$$\begin{aligned}
&\int_0^T -(\mathbf{u}_H, \partial_t \mathbf{v}_H) + (\nabla_\nu \mathbf{u}_H, \nabla_\nu \mathbf{v}_H) - (\mathbf{u}_H, (\mathbf{u} \cdot \nabla)\mathbf{v}_H) + (b(\mathbf{u}_H), \mathbf{v}_H)\,\mathrm{d}t \\
&+ \int_0^T -(C, \partial_t \tilde{C}) - (\mathbf{u}C, \nabla \tilde{C}) + K_2(\partial_2 C, \partial_2 \tilde{C}) + K_3(\partial_3 C, \partial_3 \tilde{C})\,\mathrm{d}t = \\
&= -(\mathbf{u}_{0H}, \mathbf{v}_H(0)) - (C(0), \tilde{C}(0)) + \int_0^T (S, \tilde{C})\,\mathrm{d}t
\end{aligned} \tag{20}$$

for all $(\mathbf{v}, \tilde{C})$ with $\mathbf{v} = (\mathbf{v}_H, v_3) \in H^1(0, T, \mathbf{W})$, with $v_H(T) = 0$ and $\partial_3 \mathbf{v}_H \in L^\infty(0, T; L^3(\Omega)^2)$ and $\tilde{C}$ with $\tilde{C} \in L^2(0, T, H^1)$, $\tilde{C} \in H^1(0, T, L^2)$, $\tilde{C}(T) = 0$, and $\tilde{C}$ being continuous in $\mathbf{x}_s$.

Note that the term $\int_0^T (S, \tilde{C})\,\mathrm{d}t$ makes sense since in this case $S$ denotes the delta distribution and we have

$$(S, \tilde{C}) = \int_\Omega \delta(\mathbf{x} - \mathbf{x}_s)\tilde{C}(\mathbf{x})\,\mathrm{d}\mathbf{x} = \tilde{C}(\mathbf{x}_s),$$

using the basic properties of $\delta$ and the continuity of $\tilde{C}$ in $x_s$.

**Remark.** *If $S$ is chosen to be different from the limit of the Gaussian nascent delta function, then it is easy to see that the sufficient assumption we have to make on a general $S$ hydrostatic pollution function is that it should be in $L^2_t L^2_x$.*

### 4.2 Weak formulation of the anisotropic system

Find $\mathbf{u}^\epsilon = (\mathbf{u}_H, u_3^\epsilon) \in L^2(0,t;\mathbf{V}) \cap L^\infty(0,T;L^2(\Omega)^3)$ and $C^\epsilon$ with $C^\epsilon \in L^\infty L^2$, $C^\epsilon \in L^2(0,T;H^1)$, $C^\epsilon(0) \in L^2_x$ such that

$$
\begin{aligned}
&\int_0^T -(\mathbf{u}_H^\epsilon, \partial_t \mathbf{v}_H) + (\nabla_\nu \mathbf{u}_H^\epsilon, \nabla_\nu \mathbf{v}_H) - (\mathbf{u}_H^\epsilon, (\mathbf{u}^\epsilon \cdot \nabla)\mathbf{v}_H) + (b(\mathbf{u}_H^\epsilon), \mathbf{v}_H)\,\mathrm{d}t \\
&+ \epsilon\beta\Big[\int_0^T (u_3^\epsilon, v_1) - (u_1^\epsilon, v_3)\Big]\,\mathrm{d}t \\
&+ \epsilon^2\Big[\int_0^T -(u_3^\epsilon, \partial_t v_3) + (\mathbf{u}^\epsilon \cdot \nabla u_3^\epsilon, v_3) + (\nabla_\nu u_3^\epsilon, \nabla_\nu v_3)\,\mathrm{d}t\Big] \\
&+ \int_0^T -(C^\epsilon, \partial_t \tilde{C}^\epsilon) - (\mathbf{u}^\epsilon C^\epsilon, \nabla \tilde{C}^\epsilon) + \\
&\epsilon K_1(\partial_1 C^\epsilon, \partial_1 \tilde{C}^\epsilon) - \epsilon K_1\big[(\partial_1 C^\epsilon)\tilde{C}\big]_{\Gamma_G} + K_2(\partial_2 C^\epsilon, \partial_2 \tilde{C}^\epsilon) + K_3(\partial_3 C^\epsilon, \partial_3 \tilde{C}^\epsilon)\,\mathrm{d}t = \\
&= -(\mathbf{u}_{0H}^\epsilon, \mathbf{v}_H^\epsilon(0)) - \epsilon^2(u_{03}^\epsilon, v_3^\epsilon(0)) - (C^\epsilon(0), \tilde{C}^\epsilon(0)) + \int_0^T (S^\epsilon, \tilde{C}^\epsilon)\,\mathrm{d}t
\end{aligned}
\tag{21}
$$

for all $\mathbf{v} = (\mathbf{v}_H, v_3) \in H^1(0,T;\mathbf{V})$ with $\mathbf{v}(T) = 0$ and $\tilde{C}$ such that $\tilde{C} \in H^1(0,T,H^1)$ and $\tilde{C}(T) = 0$.

Note that in this case the source term is a nascent delta function, the Gaussian pulse is in $L^\infty_x(\Omega)$, so we do not have any problems with the source term. If we used a source function of a different and more general form, it needs to satisfy the $L^2L^2$ boundedness criteria for the weak formulation to make sense.

To pass to the limit in the terms that do not include pollution in the anisotropic equation has already been worked out in the original article. What we need to work on is how to legitimately pass to the limit with the terms including the new function $C^\epsilon$. To do this our first key tool will be the energy inequality. Once we have a priori estimates for the functions in the linear terms, we can pass to the limit with those without particular problems; the challenging part will be to prove convergence for the $(\mathbf{u}^\epsilon C^\epsilon, \nabla \tilde{C}^\epsilon)$ term, here we will need a stronger result than just the weak convergence of the present functions in the product.

## 5 Proof of the main theorem

### 5.1 Energy inequality

Calculating the energy inequality for the merged system we arrive to

$$\frac{1}{2}(\left\|\mathbf{u}_H^\epsilon(t)\right\|_{L^2}^2+\epsilon^2\left\|u_3^\epsilon(t)\right\|_{L^2}^2+\left\|C^\epsilon(t)\right\|_{L^2}^2)+\int_0^t\{\|\nabla_\nu\mathbf{u}_H^\epsilon\|_{L^2}^2+\epsilon^2\|\nabla_\nu u_3^\epsilon\|_{L^2}^2\}\,\mathrm{d}\tau+$$
$$\int_0^t\{\epsilon K_1\|\partial_1C^\epsilon\|_{L^2}^2+K_2\|\partial_2C^\epsilon\|_{L^2}^2+K_3\|\partial_3C^\epsilon\|_{L^2}^2\}d\tau\le\int_0^t(S^\epsilon,C^\epsilon)\,\mathrm{d}\tau+$$
$$\frac{1}{2}(\left\|\mathbf{u}_H^\epsilon(0)\right\|_{L^2}^2+\epsilon^2\left\|u_3^\epsilon(0)\right\|_{L^2}^2+\left\|C^\epsilon(0)\right\|_{L^2}^2)+\int_0^t\epsilon K_1\langle\partial_1C^\epsilon,C^\epsilon\rangle_{\Gamma_G}\,\mathrm{d}\tau \tag{22}$$

In order to use the energy inequality to extract a priori estimates, we need to know that the right hand side of the inequality is bounded. Since the $S^\epsilon$ source term is in $L_x^\infty(\Omega)$, the only term we need to work on further is $\int_0^t\epsilon K_1\langle\partial_1C^\epsilon,C^\epsilon\rangle_{\Gamma_G}d\tau$.

We can estimate this term in the following way.

$$\begin{aligned}&\int_0^t\epsilon K_1\langle\partial_1C^\epsilon,C^\epsilon\rangle_{\Gamma_G}\,\mathrm{d}\tau\le K_1\int_0^t\sqrt{\epsilon}\|\partial_1C^\epsilon\|_{L^2(\Gamma_G)}\sqrt{\epsilon}\|C^\epsilon\|_{L^2(\Gamma_G)}\le\\&\frac{1}{\delta}K_1\frac{\epsilon}{2}\int_0^t\|\partial_1C^\epsilon\|_{L^2(\Gamma_G)}^2+\delta K_1\int_0^t\frac{\epsilon}{2}\|C^\epsilon\|_{L^2(\Gamma_G)}^2\le\\&B.T.+\delta\frac{\epsilon}{2}K_1c_T\int_0^t\|C^\epsilon\|_{W^{1,2}(\Omega)}^2,\end{aligned} \tag{23}$$

where $c_T$ is the constant provided by the trace theorem, and $B.T.$ denotes the boundary term which is bounded because of the a priori assumptions made on boundary conditions. Now we can carry on with the estimate by using the $W^{1,2}$ norm definition, exploit the fact that we are on a bounded domain and then applying the Sobolev inequality (for $n=3,p=2,p*=6$, denoting its constant by $c_S$), so we have

$$\begin{aligned}&\int_0^t\epsilon K_1\langle\partial_1C^\epsilon,C^\epsilon\rangle_{\Gamma_G}\,\mathrm{d}\tau\le\\&B.T.+\delta\frac{\epsilon}{2}K_1c_T\int_0^t\{\|C^\epsilon\|_{L^2(\Omega)}^2+\|\nabla C^\epsilon\|_{L^2(\Omega)}^2\}\,\mathrm{d}\tau\le\\&B.T.+\delta\frac{\epsilon}{2}K_1c_T\int_0^t\{\|C^\epsilon\|_{L^6(\Omega)}^2+\|\nabla C^\epsilon\|_{L^2(\Omega)}^2\}\,\mathrm{d}\tau\le\\&B.T.+\delta\frac{\epsilon}{2}K_1c_T\int_0^t\{c_S\|\nabla C^\epsilon\|_{L^2(\Omega)}^2+\|\nabla C^\epsilon\|_{L^2(\Omega)}^2\}\,\mathrm{d}\tau=\\&B.T.+\delta\frac{\epsilon}{2}K_1c_T(c_S+1)\int_0^t\|\nabla C^\epsilon\|_{L^2(\Omega)}^2\,\mathrm{d}\tau\end{aligned} \tag{24}$$

Which means that the energy inequality takes the form of

$$
\begin{aligned}
&\frac{1}{2}(\|\mathbf{u}_H^\epsilon(t)\|_{L^2}^2 + \epsilon^2\|u_3^\epsilon(t)\|_{L^2}^2 + \|C^\epsilon(t)\|_{L^2}^2) + \int_0^t \{\|\nabla_\nu \mathbf{u}_H^\epsilon\|_{L^2}^2 + \epsilon^2\|\nabla_\nu u_3^\epsilon\|_{L^2}^2\}\, \mathrm{d}\tau + \\
&\int_0^t \{(\epsilon K_1 - \delta\frac{\epsilon}{2}K_1 c_T(c_S+1))\|\partial_1 C^\epsilon\|_{L^2}^2 + (K_2 - \delta\frac{\epsilon}{2}K_1 c_T(c_S+1))\|\partial_2 C^\epsilon\|_{L^2}^2 + \\
&+ (K_3 - \delta\frac{\epsilon}{2}K_1 c_T(c_S+1))\|\partial_3 C^\epsilon\|_{L^2}^2\}\, \mathrm{d}\tau \le \frac{1}{\delta}K_1\frac{\epsilon}{2}\int_0^t \|\partial_1 C^\epsilon\|_{L^2(\Gamma_G)}^2 + \\
&+ \frac{1}{2}(\|\mathbf{u}_H^\epsilon(0)\|_{L^2}^2 + \epsilon^2\|u_3^\epsilon(0)\|_{L^2}^2 + \|C^\epsilon(0)\|_{L^2}^2) + \int_0^t (S^\epsilon, C^\epsilon)\, \mathrm{d}\tau
\end{aligned} \tag{25}
$$

The right hand side of the energy inequality is now apparently bounded, and it is easy to see that the coefficients of the pollution terms are positive, since the $\delta$ constant of the generalized Young inequality can be freely chosen in a way such that the requirements $\epsilon K_1(1-\delta\frac{1}{2}c_T(c_S+1)) > 0, (K_2-\delta\frac{\epsilon}{2}K_1 c_T(c_S+1)) > 0, (K_3 - \delta\frac{\epsilon}{2}K_1 c_T(c_S+1)) > 0$ are all satisfied.

## 5.2 A priori estimates and weak convergence

Now we can extract a priori estimates from the final form of the energy inequality. We have

- $L_t^\infty L_x^2$ boundedness for $u_1^\epsilon, u_2^\epsilon, \epsilon u_3^\epsilon$,
- $L_t^2 H_x^1$ boundedness for $u_1^\epsilon, u_2^\epsilon, \epsilon u_3^\epsilon$,
- $L_t^\infty L_x^2$ boundedness for $C^\epsilon$,
- $L_t^2 L_x^2$ boundedness for $\sqrt{\epsilon}\partial_1 C^\epsilon, \partial_2 C^\epsilon, \partial_3 C^\epsilon$.

Because of the boundedness we have weak convergence, i.e. there are subsequences still denoted by the same way, for which we have $C^\epsilon \rightharpoonup C$ in $L_t^\infty L_x^2$, $\partial_2 C^\epsilon \rightharpoonup \partial_2 C$ and $\partial_3 C^\epsilon \rightharpoonup \partial_3 C$ in $L_t^2 L_x^2$, and $\sqrt{\epsilon}\partial_1 C^\epsilon \rightharpoonup e$ for some limit element $e$ in $L_t^2 L_x^2$.

## 5.3 Passing to the limit

Passing to the limit is done without problems for the terms that do not include the pollution function, since that had been worked out before. In the linear terms the weak convergence guaranteed by the a priori estimates will be sufficient. In the nonlinear term we need an additional result that provides a weak convergence of the product of functions for which we only have weak convergence separately.

For the linear terms we can directly apply the weak convergence results of the previous section.

(i) The convergence of the $(C^\epsilon, \frac{\partial \tilde{C}}{\partial t})$ term is easy to see, since we have $C^\epsilon \rightharpoonup C$ in $L_t^\infty L_x^2$.

(ii) For the next term we have $\epsilon K_1 \int_0^T (\partial_1 C^\epsilon, \partial_1 \tilde{C}) = \sqrt{\epsilon} K_1 \int_0^T (\sqrt{\epsilon}\partial_1 C^\epsilon, \partial_1 \tilde{C})$, which goes to 0 as $\epsilon \to 0$ because of the Cauchy-Schwartz inequality, since we have $\sqrt{\epsilon}\partial_1 C^\epsilon$ bounded in $L_t^2 L_x^2$, and $\tilde{C}$ is a test function.

(iii) As we have seen before, we have $\partial_2 C^\epsilon \rightharpoonup \partial_2 C$ in $L_t^2 L_x^2$, which by definition provides that

$$\int_0^T (\partial_2 C^\epsilon, f) \mathrm{dt} \to \int_0^T (\partial_2 C, f)\, \mathrm{d}t \qquad \forall f \in L_t^2 L_x^2.$$

(iv) Analogously,

$$\int_0^T (\partial_3 C^\epsilon, f) \mathrm{dt} \to \int_0^T (\partial_3 C, f)\, \mathrm{d}t \qquad \forall f \in L_t^2 L_x^2.$$

(v) The $\epsilon K_1 \int_0^T \left[(\partial_1 C^\epsilon)\tilde{C}\right]_{\Gamma_G} \mathrm{d}t$ line integral vanishes as $\epsilon \to 0$ since by the Cauchy-Schwartz equation we can estimate the integral by two bounded terms ($\partial_1 C^\epsilon$ was assumed to be finite in norm on the lower boundary).

(vi) To prove $(S^\epsilon, \tilde{C}) \to (S, \tilde{C})$ we need to see that $\lim_{\epsilon\to 0} \int_\Omega S^\epsilon(\mathbf{x}, \mathbf{x}_s) \cdot \tilde{C}(\mathbf{x})\, \mathrm{d}\mathbf{x} \to \tilde{C}(\mathbf{x}_s)$. Using the substituation $\mathbf{y} = (\mathbf{x} - \mathbf{x}_s)/\epsilon$ we have

$$\int_\Omega \gamma \epsilon^{-1} e^{\frac{-(\mathbf{x}-\mathbf{x}_s)^2}{\epsilon^2}} \tilde{C}(\mathbf{x})\, \mathrm{d}\mathbf{x} = \int_{\Omega_\mathbf{y}} \gamma e^{-\mathbf{y}^2} \tilde{C}(\epsilon \mathbf{y} + \mathbf{x}_s)\, \mathrm{d}\mathbf{y}$$

This term can be dominated by $\|\tilde{C}\|_\infty \cdot \int_{\Omega_\mathbf{y}} \gamma e^{-\mathbf{y}^2} \tilde{C}\, \mathrm{d}\mathbf{y} = \|\tilde{C}\|_\infty$, so we can apply the dominated convergence theorem using that the test function $\tilde{C} \in L_t^2 L_x^\infty$ and arrive to

$$\lim_{\epsilon\to 0} (S^\epsilon, \tilde{C}) = \int_{\Omega_\mathbf{y}} \lim_{\epsilon\to 0} \gamma e^{-\mathbf{y}^2} \tilde{C}(\epsilon \mathbf{y} + \mathbf{x}_s)\, \mathrm{d}\mathbf{y} = \int_{\Omega_\mathbf{y}} \gamma e^{-\mathbf{y}^2} \tilde{C}(\mathbf{x}_s)\, \mathrm{d}\mathbf{y} = \tilde{C}(\mathbf{x}_s),$$

using the continuity of $\tilde{C}$ in $\mathbf{x}_s$.

As for the nonlinear term $(\mathbf{u}^\epsilon C^\epsilon, \nabla \tilde{C})$, we need some sort of compactness theorem to handle it.

**Remark.** *As a matter of fact, we actually have three terms here, namely $(u_i^\epsilon C^\epsilon, \partial_i \tilde{C})$ for $i = 1, 2, 3$, and for the values $1, 2$ of $i$ the weak convergence of the $u_i^\epsilon C^\epsilon$ product is given by the fact that, as it is proved in the reference article, we actually have strong convergence for the horizontal velocities, and because of the a priori estimate from the energy bound we know we have weak* *convergence for the $C^\epsilon$ term. But this argument does not really solve the challenge caused by nonlinearity because it does not hold for $u_3$, so we still need an additional compactness theorem as mentioned before.*

In this case the Aubin-Lions lemma can not be applied since it would require for $C^\epsilon$ to be in a compactly embedded space such as $L^2H^1$, which is not the case here because we only know boundedness results for $\sqrt{\epsilon}\partial_1 C$, but not for $\partial_1 C$.

We will show that the following lemma from [Lions, 1996] is applicable in our case.

**Lemma 1.** *Let $g_n$, $h_n$ converge weakly to $g, h$ respectively in $L^{p_1}(0,T;L^{p_2}(\Omega))$, $L^{q_1}(0,T;L^{q_2}(\Omega))$ where $1 \le p_1, p_2 \le +\infty$,*

$$\frac{1}{p_1} + \frac{1}{q_1} = \frac{1}{p_2} + \frac{1}{q_2} = 1.$$

*We assume in addition that*

- $\frac{\partial g^n}{\partial t}$ *is bounded in $L^1(0,T;W^{-m,1}(\Omega))$ for some $m \ge 0$ independent of $n$.*
- $\left\|h^n - h^n(\cdot + \xi, t)\right\|_{L^{q_1}(0,T;L^{q_2}(\Omega))} \to 0$ *as $|\xi| \to 0$, uniformly in $n$.*

*Then, $g^n h^n$ converges to $gh$ (in the sense of distributions on $\Omega \times (0,T)$).*

What we will show in the following is that for the functions $u_i^\epsilon$ and $C^\epsilon$ the assumptions of Lemma 1 are satisfied, which means that $u_i^\epsilon C^\epsilon \rightharpoonup u_i C$ holds.

**Lemma 2.** *We have $u_i^\epsilon C^\epsilon \rightharpoonup u_i C$.*

*Proof.* We have seen from the a priori estimates that $C^\epsilon \rightharpoonup C$ in $L_t^\infty L_x^2$, and we know from the reference article that $u_i^\epsilon \rightharpoonup u_i$ in $L_t^2 L_x^2$.

Let us choose $q_1 = q_2 = p_1 = p_2 = 2$. This is fine since we are on a finite domain, so $C^\epsilon \in L^\infty L^2$ implies $C^\epsilon \in L^2 L^2$.

- **Bound for the time derivative**.

  $$\frac{\partial C^\epsilon}{\partial t} = -u^\epsilon \nabla C^\epsilon + \sqrt{\epsilon} K_1 \partial_1 \partial_1 \sqrt{\epsilon} C^\epsilon + K_2 \partial_2 \partial_2 C^\epsilon + K_3 \partial_3 \partial_3 C^\epsilon - S^\epsilon$$

  On one hand, the $S^\epsilon$ Gaussian source term is smooth, and we see that the second part of the right hand side is in $W^{-1,2}$, simply using that $\partial_1 \sqrt{\epsilon} C^\epsilon, \partial_2 C^\epsilon, \partial_3 C^\epsilon$ are in $L^2$. On the other hand, the first part is in $W^{-1,1}$, because we have

  $$u^\epsilon \nabla C^\epsilon = \nabla \cdot (u^\epsilon C^\epsilon) - \nabla \cdot (u^\epsilon) C,$$

  and we just have to use that the divergence of $u^\epsilon$ is zero and $u^\epsilon C^\epsilon$ is in $L_x^1$.

  Eventually we have $\frac{\partial C}{\partial t} \in L^1 W^{-1,1}$.

- **Equicontinuity**. We need to show that $\left\|u_i^n - u_i^n(\cdot+\xi,t)\right\|_{L^2L^2} \to 0$ as $\xi \to 0$. Since in the time domain the condition naturally holds, we will focus on the space part of the proof.

  First we show that for a function $f$ in the function space $C_0^\infty$ the condition holds.

$$\left\|f(x+\xi)-f(x)\right\|_{L^2} = \left(\int_\Omega \left|f(x+\xi)-f(x)\right|^2 dx\right)^{1/2} \leq$$

$$\left(\int_\Omega (\sup_x \left|f(x+\xi)-f(x)\right|)^2 d\mathrm{x}\right)^{1/2} = \left\|f(x+\xi)-f(x)\right\|_{L^\infty} \cdot |\Omega|^{1/p}.$$

  So we have

$$\lim_{\xi\to 0}\left\|f(x+\xi)-f(x)\right\|_{L^2} \leq |\Omega|^{1/p} \cdot \lim_{\xi\to 0}(\sup_x \left|f(x+\xi)-f(x)\right|) = 0,$$

  since $f \in C_0^\infty$.

  As a second step we show that the condition holds for any function $f \in L^2$. Here we exploit the fact that $C_0^\infty$ is dense in the $L^p$ function space. By definition this means that $\exists \varphi_n \in C_0^\infty(\Omega)$ such that $\varphi_n \to f$ in $L^p(\Omega)$. With this we have

$$\left\|f(x+\xi)-f(x)\right\|_{L^2} =$$

$$\left\|f(x+\xi)-\varphi_n(x+\xi)+\varphi_n(x+\xi)-\varphi_n(x)+\varphi_n(x)-f(x)\right\|_{L^2} \leq$$

$$\left\|f(x+\xi)-\varphi_n(x+\xi)\right\|_{L^2} + \left\|\varphi_n(x+\xi)-\varphi_n(x)\right\|_{L^2} + \left\|\varphi_n(x)-f(x)\right\|_{L^2}$$

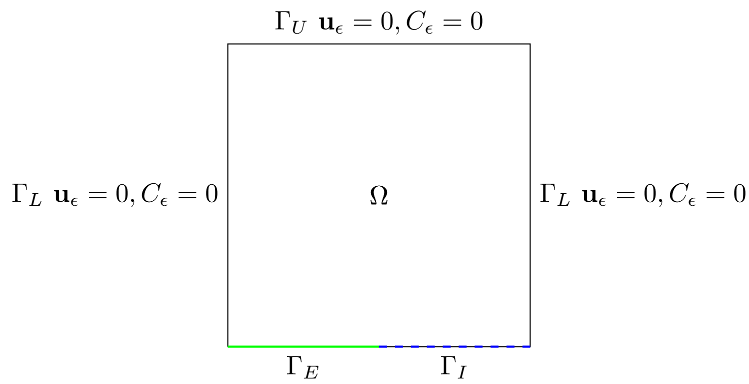
  The first and third term goes to 0 as $\xi \to 0$, since $\varphi_n \to f$ in $L^p(\Omega)$. On the other hand in the first part of the proof we have shown exactly that the second term goes to 0. Which proves that $\left\|u_i^n - u_i^n(\cdot+\xi,t)\right\|_{L^2L^2} \to 0$ as $\xi \to 0$. Eventually this means that we can apply the lemma with $h^n = u_i^\epsilon, g^n = C^\epsilon$, which proves the statement of the lemma.

$\square$

# 6 A generalized model

We now will present the convection dominated coastal model, which is a generalization of the previous inland model by adapting the model for seaside conditions.

On the ground layer we naturally separate the boundary into two different parts in this model, $\Gamma_G = \Gamma_E \bigcup \Gamma_I$; $\Gamma_E$ denoting the part of the boundary above the ground, and $\Gamma_I$ denoting the section that is interacting with the sea. The




separation is a sort of expansion or generalization of the ground boundary — while above ground parts it is reasonable to use no-slip conditions, above the ocean it is natural to assume that the wind has an interaction with the waves, there is no such rigidness as in the ground case. For this cause we will include $\theta$, which represents the wind traction on the ocean surface. Analogously, we need to make similar changes for the pollution concentration as well, above sea territory it is not natural to use $\partial_3 C = 0$, since the particles can now sink under the boundary level.

On $\Gamma_E$ we have the same boundary conditions as in the inland case, $\mathbf{u}^\epsilon = 0, \partial_3 C^\epsilon = 0, \partial_2 C^\epsilon = 0, \partial_1 C^\epsilon$ bounded. On $\Gamma_I$ on the other hand we have $\nu_3 \partial_3 \mathbf{u}^\epsilon_H = \theta_H, u_3 = 0$ with assuming a finite wind energy, i.e. the functions $\theta$ being in $L^2(0,T;H^{-1/2}(\Gamma_I))$. For the pollution concentration we use the same approach and assume that the functions $\partial_1 C^\epsilon, \partial_2 C^\epsilon, \partial_3 C^\epsilon$ are equal to some sufficiently regular, $\epsilon$-independent functions defined on the respective boundaries. More accurately, we assume that we have a $C^\epsilon_{\Gamma_G} = \sigma$ sediment function on the lower boundary, where $\sigma \in L^2(0,T;H^1(\Gamma_G))$. If the partial derivatives of $\sigma$ are denoted by $\sigma_1, \sigma_2, \sigma_3$, we have $\partial_1 C^\epsilon = \sigma_1$ on $\Gamma_G$, and $\partial_2 C^\epsilon = \sigma_2$, $\partial_3 C^\epsilon = \sigma_3$ on $\Gamma_I$. (Of course $\sigma$ has to be a function such that we have $\sigma_2$ and $\sigma_3$ equal to zero on the $\Gamma_E$ part).

We will discuss in more details the parts that are different from the inland case.

The weak formulation of the coastal generalized systems will be different from the original inland version in that *a)* it will contain a boundary term for the wind because of the air - ocean interface, and *b)* it will contain boundary terms for $\partial_2 C, \partial_3 C$, which had been assumed to be zero before on the ground.

The weak form of the anisotropic system is the following.

$$
\begin{aligned}
&\int_0^T -(\mathbf{u}_H^\epsilon, \partial_t \mathbf{v}_H) + (\nabla_\nu \mathbf{u}_H^\epsilon, \nabla_\nu \mathbf{v}_H) - (\mathbf{u}_H^\epsilon, (\mathbf{u}^\epsilon \cdot \nabla)\mathbf{v}_H) + (b(\mathbf{u}_H^\epsilon), \mathbf{v}_H)\,\mathrm{d}t \\
&+ \epsilon\beta\big[\int_0^T (u_3^\epsilon, v_1) - (u_1^\epsilon, v_3)\big]\,\mathrm{d}t \\
&+ \epsilon^2\big[\int_0^T -(u_3^\epsilon, \partial_t v_3) + (\mathbf{u}^\epsilon \cdot \nabla u_3^\epsilon, v_3) + (\nabla_\nu u_3^\epsilon, \nabla_\nu v_3)\,\mathrm{d}t\big] \\
&+ \int_0^T -(C^\epsilon, \partial_t \tilde{C}^\epsilon) - (\mathbf{u}^\epsilon C^\epsilon, \nabla \tilde{C}^\epsilon) + \epsilon K_1(\partial_1 C^\epsilon, \partial_1 \tilde{C}^\epsilon) - \epsilon K_1\langle \sigma_1, \tilde{C}\rangle_{\Gamma_G} + \\
&K_2(\partial_2 C^\epsilon, \partial_2 \tilde{C}^\epsilon) - K_2\langle \sigma_2, \tilde{C}\rangle_{\Gamma_I} + K_3(\partial_3 C^\epsilon, \partial_3 \tilde{C}^\epsilon) - K_3\langle \sigma_3, \tilde{C}\rangle_{\Gamma_I}\,\mathrm{d}t = \\
&= -(\mathbf{u}_{0H}^\epsilon, \mathbf{v}_H^\epsilon(0)) - \int_0^T \langle \theta_H, \mathbf{v}_H\rangle_{\Gamma_I} - \epsilon^2(u_{03}^\epsilon, v_3^\epsilon(0)) - (C^\epsilon(0), \tilde{C}(0)) + \int_0^T (S^\epsilon, \tilde{C})\,\mathrm{d}t
\end{aligned}
\tag{26}
$$

The weak version of the hydrostatic equations:

$$
\begin{aligned}
&\int_0^T -(\mathbf{u}_H, \partial_t \mathbf{v}_H) + (\nabla_\nu \mathbf{u}_H, \nabla_\nu \mathbf{v}_H) - (\mathbf{u}_H, (\mathbf{u} \cdot \nabla)\mathbf{v}_H) + (b(\mathbf{u}_H), \mathbf{v}_H)\,\mathrm{d}t \\
&+ \int_0^T -(C, \partial_t \tilde{C}) - (\mathbf{u}C, \nabla \tilde{C}) + \\
&K_2(\partial_2 C, \partial_2 \tilde{C}) - K_2\langle \sigma_2, \tilde{C}\rangle_{\Gamma_I} + K_3(\partial_3 C, \partial_3 \tilde{C}) - K_3\langle \sigma_3, \tilde{C}\rangle_{\Gamma_I}\,\mathrm{d}t = \\
&= -(\mathbf{u}_{0H}, \mathbf{v}_H(0)) - \int_0^T \langle \theta_H, \mathbf{v}_H\rangle_{\Gamma_I} - (C(0), \tilde{C}(0)) + \int_0^T (S, \tilde{C})\,\mathrm{d}t
\end{aligned}
\tag{27}
$$

The analogous terms that we saw in the weak formulation will be different in the energy inequality as well. The coastal, generalized version of the energy inequality becomes the following.

$$
\begin{aligned}
&\frac{1}{2}(\|\mathbf{u}_H^\epsilon(t)\|_{L^2}^2 + \epsilon^2\|u_3^\epsilon(t)\|_{L^2}^2 + \|C^\epsilon(t)\|_{L^2}^2) + \int_0^t \{\|\nabla_\nu \mathbf{u}_H^\epsilon\|_{L^2}^2 + \epsilon^2\|\nabla_\nu u_3^\epsilon\|_{L^2}^2\}\,\mathrm{d}\tau + \\
&\int_0^t \{\epsilon K_1\|\partial_1 C^\epsilon\|_{L^2}^2 + K_2\|\partial_2 C^\epsilon\|_{L^2}^2 + K_3\|\partial_3 C^\epsilon\|_{L^2}^2\}\,\mathrm{d}\tau \leq \\
&\frac{1}{2}(\|\mathbf{u}_H^\epsilon(0)\|_{L^2}^2 + \epsilon^2\|u_3^\epsilon(0)\|_{L^2}^2 + \|C^\epsilon(0)\|_{L^2}^2) + \int_0^t (S^\epsilon, C^\epsilon)\,\mathrm{d}\tau - \int_0^t \langle \theta_H, \mathbf{u}_H\rangle_{\Gamma_I} \\
&\int_0^t \epsilon K_1\langle \sigma_1, \sigma\rangle_{\Gamma_G} + K_2\langle \sigma_2, \sigma\rangle_{\Gamma_I} + K_3\langle \sigma_3, \sigma\rangle_{\Gamma_I}\,\mathrm{d}\tau
\end{aligned}
\tag{28}
$$

Since the $\sigma$ boundary function is in $L^2H^1$, we have boundedness on the right hand side. Even though by first look the inequality seems a bit different from the inland one, it provides the same estimates and weak convergences. We have

- $L^\infty_t L^2_x$ boundedness for $u^\epsilon_1, u^\epsilon_2, \epsilon u^\epsilon_3$,
- $L^2_t H^1_x$ boundedness for $u^\epsilon_1, u^\epsilon_2, \epsilon u^\epsilon_3$,
- $L^\infty_t L^2_x$ boundedness for $C^\epsilon$,
- $L^2_t L^2_x$ boundedness for $\sqrt{\epsilon}\partial_1 C^\epsilon, \partial_2 C^\epsilon, \partial_3 C^\epsilon$.

As for passing to the limit as $\epsilon \to 0$ in the coastal version of the anisotropic system seen above, there is no mathematical novelty compared to the inland case.

**Remark.** *If we switch to the "informationless", fix coordinate system that does not match the downwind direction and we drop the assumption on the flow being convection dominated ($\epsilon$ - diffusion), then we can relax the restrictions on $C^\epsilon$ on the boundary. This is true because in this case we do not have a disappearing $\epsilon$ term in front of the $K_1$ constant, and we can use the analogous estimate as in the inland case at the energy inequality and arrive to having a $(K_1 - \delta\frac{K_1+K_2+K_3}{2}c_T(c_S+1))\|\partial_1 C^\epsilon\|^2_{L^2}$ term with a positive coefficient on the left hand side and a bounded $\frac{1}{\delta_i}\frac{K_i}{2}\int_0^t \|\partial_i C^\epsilon\|^2_{L^2(\Gamma_G)}$ integral on the right hand side — the only assumption we need is $\partial_i C^\epsilon$ to be in $L^2_t L^2_x$ on the boundary.*

# 7 Concluding remarks

**Remark.** In the paper of [Temam and Ziane, 2005], on p35 it is suggested that the domain we work on has the form of $p_0 < p < p_1$, because for the thin portion of air between the earth and the isobar $p = p_1$, another specific simplified model would be necessary. (However other authors use the $z = 0$ simplified ground level cut as well, such as [Hosseini, 2013]) This means that a way to make the model more precise is to cut the domain at an isobar a little bit off the physical ground. This would of course radically change the boundary conditions we use for the functions (in fact it is not trivial how to change them for this case), because it would take away the advantage of having a wall-like, natural and physical boundary condition for our domain to support it. If we are positively above the ground, $\partial_3 C = 0$ would represent a strange reflective layer.

**Remark.** Note that the aspect ratio going to zero is not only legitimate on a global scale. When we switch from a global model to a local one in order to be able to use the beta plane approximation, we still have hundreds of thousands square kilometers versus 1—5 kilometers.

**Remark.** The [Temam and Ziane, 2005] article works with the beta plane approximation, which is appropriate for example for midlatitude regional studies, where the curvature is minimal. If we take this as the starting idea, and consider a local part of the ocean, we will most likely not have genuine physical lateral boundaries for the ocean water in the model, such as would be a coast for example. Nevertheless what the article uses as a lateral boundary condition is the salt flux being zero, which represents a wall-like boundary. What would possibly be more natural is the salt flux being constant, but it would naturally make things mathematically more complicated on the other hand. This is another example of how challenging it can be to find the right boundary terms when we are handling OBCs.

**Remark.** We chose the Gaussian nascent delta function to give shape to the source term, but it can be defined in several different ways using the many different pulses that approximate the delta function (choosing only those however that are physically meaningful in our case, for example its values are nonnegative), e.g.

a) the unit impulse: $\delta_\epsilon(\mathbf{x}-\mathbf{x_s}) = \epsilon/2, \left|(\mathbf{x}-\mathbf{x_s})\right| < 1/\epsilon$,

b) the Lorentzian pulse: $\delta_\epsilon(\mathbf{x}-\mathbf{x_s}) = \epsilon/(1+\pi^2\epsilon^2(\mathbf{x}-\mathbf{x_s})^2)$.

The Gaussian and Lorentzian version of the source term is continuous and bounded on $\Omega$, it is actually in $L^\infty$ in space (and in $L^2$ as a consequence), while it is easy to see that the piecewise constant unit pulse is in $L^2$ as well. Since the time dependence of this source is described by the Heaviside function, the time norm is also finite.

# A The existence of weak solution of the hydrostatic system of the polluted atmosphere

What we have showed is that the weak solutions of the anisotropic system converge to the weak solution of the hydrostatic system of the polluted atmosphere. Without having an existence theorem for the hydrostatic equations, the weak solutions of the anisotropic system could just be converging to the empty set. The reference article for unpolluted fluids states only the convergence part as well, but the existence is implicitly assumed since it is a well-known result.

To complete our work of expanding the original result to the case of a fluid containing a certain pollution concentration, we also need an existence theorem.

The idea is that the polluted atmosphere can be viewed as the air-analogous version of the salty sea water in terms of the equations that represent it. In both cases we have a fluid which can be seen approximately as a homogenous incompressible fluid carrying some other particles in a significant concentration, the basis fluid motion is well described by the incompressible Navier-Stokes equations, and the additional concentration is described by an advection-diffusion equation.

The paper of [Temam and Ziane, 2005] provides an existence result for an arbitrary time interval $[0, t_1]$ for the system describing the salty ocean water with temperature. The structure of the equations themselves describing the system is essentially the same, if we drop the equation describing the temperature, we basically arrive to the same system that describes the polluted atmosphere.

The differences are the following:

a) Boundary conditions: On every boundary they use a zero flux for the salinity, $\frac{\partial S}{\partial \mathbf{n}_s} = 0$. This coincides with our lower boundary conditions, but on the lateral boundary we use a fix constant.

b) Spatial weak formulation: In the [Temam and Ziane, 2005] article they use spatial weak formulation.

c) The $K_1$ term: In the pollution equation the $K_1$ term is missing in our version of the concentration equation.

d) Concentration mean integral: for the salinity they use a mean integral equals zero condition after a shift, which is not a big constriction, since the shift can be reversed. (This is something that we use as well anyway, since we subtracted the background flow when discussing the boundary conditions.)

e) The $K_C$ term: In order to have coercivity, in the weak formulation for the concentration part we multiply by $K_C\tilde{C}$, not just by $\tilde{C}$.

Remember that the weak formulation that we need an existence theorem for is

$$\begin{aligned}
&\int_0^T -(\mathbf{u}_H, \partial_t \mathbf{v}_H) + (\nabla_\nu \mathbf{u}_H, \nabla_\nu \mathbf{v}_H) - (\mathbf{u}_H, (\mathbf{u}\cdot\nabla)\mathbf{v}_H) + (b(\mathbf{u}_H), \mathbf{v}_H)\,\mathrm{d}t \\
&+ \int_0^T -(C, \partial_t \tilde{C}) - (\mathbf{u}C, \nabla\tilde{C}) + K_2(\partial_2 C, \partial_2 \tilde{C}) + K_3(\partial_3 C, \partial_3 \tilde{C})\,\mathrm{d}t = \\
&= -(\mathbf{u}_{0H}, \mathbf{v}_H(0)) - (C(0), \tilde{C}(0)) + \int_0^T (S, \tilde{C})\,\mathrm{d}t
\end{aligned} \tag{29}$$

While the equation the article provides the result for has the form of

$$\big(\frac{\mathrm{d}}{\mathrm{d}t}U, \tilde{U}\big)_H + a(U, \tilde{U}) + b(U, U, \tilde{U}) + e(U, \tilde{U}) = l(\tilde{U}),$$

where the definition of the above terms can be found on page 21 in the article of [Temam and Ziane, 2005]. In this version the weak formulation is only performed in space, and the space derivatives are not in every case translated to the text function when it is possible.

What the above equation means for example for the salinity is

$$\begin{aligned}
&\int_{\mathcal{M}} K_S \frac{\mathrm{d}}{\mathrm{d}t} S \cdot \tilde{S} d\mathcal{M} + K_S \int_{\mathcal{M}} \big(\mu_s \nabla S \nabla \tilde{S} + \nu_S \frac{\partial S}{\partial z}\frac{\partial \tilde{S}}{\partial z}\big)\,\mathrm{d}\mathcal{M} + \\
&+ K_S \int_{\mathcal{M}} \big(v\nabla S + w\frac{\partial S}{\partial z}\big)\tilde{S}\,\mathrm{d}\mathcal{M} = K_S \int_{\mathcal{M}} F_S \tilde{S}\,\mathrm{d}\mathcal{M} + \big(K_S \mu_S \frac{\partial S}{\partial x}\tilde{S}\big)_{\Gamma_i}.
\end{aligned}$$

**Step 1.** The first step is to redefine the norms so that they match our system.

$$((U,\tilde{U})) = ((v,\tilde{v}))_1 + K_C((C,\tilde{C}))_2$$

$$((v,\tilde{v}))_1 = \int_\Omega \nabla_\nu v \nabla_\nu \tilde{v} \, \mathrm{d}\omega$$

$$((C,\tilde{C}))_2 = \int_\Omega K_2 \partial_2 C \partial_2 \tilde{C} + K_3 \partial_3 C \partial_3 \tilde{C} \, \mathrm{d}\omega$$

$$\|U\| = ((U,U))^{1/2}$$

$$(U,\tilde{U})_H = \int_\Omega -v\tilde{v} - K_C C\tilde{C} \, \mathrm{d}\omega$$

$$|U|_H = \left|(U,U)_H\right|^{1/2}$$

**Step 2.** With this, now the weak formulation of the polluted atmosphere can be brought to an analogous format as seen above:

$$\int_0^T (U, \frac{\mathrm{d}}{\mathrm{d}t}\tilde{U})_H \, \mathrm{d}t + \int_0^T a(U,\tilde{U}) + b(U,U,\tilde{U}) + e(U,\tilde{U}) \, \mathrm{d}t =$$

$$= \int_0^T l(\tilde{U}) \, \mathrm{d}t + \int_0^T \int_\Omega U_0 \tilde{U}(0) d\omega \, \mathrm{d}t$$

with

$$a(U,\tilde{U}) = a_1(U,\tilde{U}) + K_C a_2(U,\tilde{U})$$

$$a_1(U,\tilde{U}) = \int_\Omega \nabla_\nu v \nabla_\nu \tilde{v} \, \mathrm{d}\omega$$

$$a_2(U,\tilde{U}) = \int_\Omega K_2 \partial_2 C \partial_2 \tilde{C} + K_3 \partial_3 C \partial_3 \tilde{C} \, \mathrm{d}\omega$$

$$b = b_1 + K_2 b_2$$

$$b_1(U,\tilde{U},U^\#) = -\int_\Omega (v\tilde{v} + v\tilde{w}) \nabla v^\#$$

$$b_2(U,\tilde{U},U^\#) = -\int_\Omega (v\tilde{C} + w\tilde{C}) \nabla C^\#$$

$$e(U,\tilde{U}) = \int_\Omega b(v)\tilde{v} \, \mathrm{d}\omega$$

$$l(\tilde{U}) = -\int_\Omega S\tilde{C} \, \mathrm{d}\omega.$$

Here the term $b(\mathbf{u}_H) = \alpha(-u_2, u_1)$ represents the cross product, and we use the fact that the vertical velocity $w$ can be expressed as $w = w(v)$ because the divergence of the three dimensional velocity is zero.

**Step 3.** The next step is to check if the properties that yield the foundation of the proof still hold. It is easy to see that $a$ is trilinear continuous, the coercivity of $a$ is also trivial, moreover $e$ is bilinear continuous and $e(U, U) = 0$.

What changes slightly is the proof of the results available on the expression $b$ since the space derivatives in our case are applied on the test function.

- To prove the bound for $U, \tilde{U} \in V, U^{\#} \in V_{(2)}$, in this case we can not use the $L_\infty$ bound on the third term, since we would eventually arrive to $\|C^{\#}\|_{H^3}$, which means that we can not close the bound that way. We can use

$$\int_\Omega w\tilde{C}\nabla C^{\#} d\omega \leq c\|w\|_{L^2} \|\tilde{C}\|_{L^4} \|\nabla C^{\#}\|_{L^4} \leq$$
$$\leq c\|U\| \|\tilde{U}\| \|\nabla C^{\#}\|_{H^1} \leq c\|U\| \|\tilde{U}\| \|U^{\#}\|_{V^{(2)}}$$

- For $U, U^{\#} \in V, \tilde{U} \in V_{(2)}$ we use

$$\int_\Omega w\tilde{C}\nabla C^{\#} d\omega \leq c\|\tilde{C}\|_{L^\infty} \|w\|_{L^2} \|\nabla C^{\#}\|_{L^2} \leq c\|U\| \|\tilde{U}\|_{V^{(2)}} \|U^{\#}\|,$$

  exploiting the fact that in $n = 3$ we have $H^2 \subset L^\infty$.

- The proof of $b(U, \tilde{U}, U^{\#}) = -b(U, U^{\#}, \tilde{U})$ can be derived similarly as we derived

$$\int_0^T (\mathbf{u}\nabla C, \tilde{C})\, \mathrm{d}t = -\int_0^T (-\mathbf{u}C, \nabla\tilde{C})\, \mathrm{d}t,$$

  using the integration by parts formula, observing that the boundary terms disappear, and finally using the fact that the divergence of the velocity.

- The establishment of the improvement of the first bound works the same way as in the original article, except that we don't apply the switch provided by the result of the previous point, i.e. we perform the estimate

$$\int_\Omega w\tilde{C}\nabla C^{\#}\, \mathrm{d}\omega \leq \|w\|_{L^2} \|\tilde{C}\|_{L^3} \|\nabla C^{\#}\|_{L^6},$$

  and after this everything follows the same way.

This means that all the basic results and properties of the functions $a, b, e$ still hold in this version.

**Step 4.** The operator form of the equation.

The change in this part of the proof is that we will need an additional operator $Z$ for the time derivative part. In the original version $\frac{\mathrm{d}U}{\mathrm{d}t}$ was used as an operator acting on $\tilde{U}$, the result defined to be their inner product. In this version, because we apply the weak formulation in space-time, the time derivation is applied on the test function, so we will need the following operator $Z$:

$$< Z(U), \tilde{U} >= \int_0^T (U, \frac{d\tilde{U}}{dt})_H \, \mathrm{d}t.$$

With this the operator form of the equation is

$$ZU + AU + B(U,U) + EU = l + U_0,$$

where all the other operators are understood in a way that the time integration is included. Since in this version we work with both time-space and only-space (stationary case) weak formulation as well, it is always assumed that we use the one according to the context.

**Step 5.** The stationary case works as it worked in the original article, since the operator form of the new version is created in a way that $ZU$ and $U_0$ contains all the time-related part, after dropping it we arrive to the same equation as in the original case.

**Step 6.** Time-dependent case. Once again we remind ourselves that in this case, since in our version we performed the weak formulation also in time, we can not use notations such as $\frac{d\tilde{U}_k}{dt}$ for the operator that will result in the inner product of the function's time derivative and the test function. Instead, we can use $Z\tilde{U}_k$, which as an operator acting on $\tilde{U}$ yields the inner product of the function and the time derivative of the test function.

Even though the time derivative here is applied on the test function, it does not make sense to write something like $\tilde{U}^n - \tilde{U}^{n-1}$, since $\tilde{U}$ is the test function. But using the time difference for the solution function actually works, and the original article arrives from the same start to a weak formulation where the derivative is applied on the test function.

Basically with the time discretization form of

$$\frac{1}{\Delta t}(U^n - U^{n-1}, \tilde{U})_H + a(U^n, U^n) + b(U^n, U^n, \tilde{U}) + e(U^n, \tilde{U}) = l(\tilde{U}),$$

we arrive to the expression

$$-\int_0^T (U, \tilde{U})_H \psi' \, \mathrm{d}t + \int_0^T \left[a(U, \tilde{U}) + b(U, U, \tilde{U}) + e(U, \tilde{U})\right] \psi \, \mathrm{d}t =$$

$$= (U_0, \tilde{U})\psi(0) + \int_0^T l(\tilde{U})\psi \, \mathrm{d}t,$$

where $\psi$ is a function defined on $[0, T]$, used to create the time-dependent version $\tilde{U}\psi$ of the test function. If we rename this product simply to be $\tilde{U}$, the time-dependent test function, we arrive exactly to,

$$-\int_0^T (U, \partial_t \tilde{U})_H \, \mathrm{d}t + \int_0^T \left[a(U, \tilde{U}) + b(U, U, \tilde{U}) + e(U, \tilde{U})\right] \mathrm{d}t =$$

$$= (U_0, \tilde{U}(0)) + \int_0^T l(\tilde{U}) \, \mathrm{d}t,$$

which is what we needed.

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